Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of in-house industry experience at HDFC Asset Management building and running production GenAI solutions end to end: department-specific RAG chatbots and knowledge-retrieval applications on Azure OpenAI and LlamaIndex with pgvector, tool-calling agents that automate analyst workflows, and the Python FastAPI services, data pipelines and ReactJS frontends around them, deployed on AWS with LDAP-based access control. Worked directly with Compliance, Client Service and Investment stakeholders in a regulated financial firm, with Compliance reviewing every application before release; acted as de facto team lead for up to 4 interns and was twice recognised internally with Best Team Project (2024, 2025). Recently completed an MSc in Data Science at the University of Bristol (result awaited), with a dissertation in partnership with AstraZeneca. Seeking an AI Engineer role delivering LLM, RAG and workflow-automation solutions with measurable business benefit.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

LLM Applications, RAG & Agentic Workflows

- Designed and built department-specific **RAG** chatbots and knowledge-retrieval applications used across Compliance, Client Service and Investment (**LlamaIndex** with **Azure OpenAI GPT-3.5 & Llama-3**), with a separate **pgvector** database and **Azure OpenAI** embeddings for each department so access stayed siloed; built end to end with a **ReactJS** frontend and **Python FastAPI** microservices.
- Built **agentic tool-calling** workflows where AI agents call **Python functions** as tools to run SQL-backed fund-comparison calculations and generate **PDF** comparison reports on user request; guardrailed responses through **system prompts** and versioned every prompt change in **Git**.
- Built a **WhatsApp** fund-comparison bot (via the **Netcore** API) used daily by the Sales team (roughly **50 to 100 users**, **400 to 1,000 requests** in working hours): **Azure OpenAI tool calling** picks the funds and comparison type; fixed failing parameter extraction by adding example inputs to the **JSON tool definitions**.
- Released each application to real users and iterated on what broke. Chose models by cost: **Llama-3 self-hosted on a GPU EC2 instance** for the largest department (Client Service) and stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models.

Data Pipelines, Automation & Integration

- Built a **Python** email-processing automation that turns over **300 daily emails** from credit rating agencies into structured data, fuzzy-matches them to tracked companies and sends analysts a daily summary, saving **60 manual analyst hours a month**; replaced **AWS Textract** with **Azure OpenAI** when a rating agency began sending emails as images, which also improved accuracy.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), integrated with legacy data from analysts' Excel files; developed **Power BI** dashboards that improved analyst efficiency by **10 hours a week**, recreated in the **ReactJS** frontend for wider access without Power BI licensing costs.

Secure Deployment, Governance & Team Delivery

- Designed the infrastructure for the GenAI applications and analytics platforms on **AWS** utilizing core services such as **EC2, S3, ECR** and **Lambda**, with **Docker** microservices (FastAPI, uvicorn), separate frontend and backend repositories and containers, and a one-click **Jenkins** and **GitLab CI/CD** pipeline for dev and prod.
- Implemented **RBAC** for the GenAI apps with a Python authentication function against the company **LDAP** server; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the **VAPT** (vulnerability assessment and penetration testing); coordinated with the security team on port whitelisting.
- Worked directly with stakeholders in **Client Service, Compliance and Investment** to turn requirements into working tools that replace manual work, with **Compliance** reviewing every application before release; acted as **de facto team lead for up to 4 interns** in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: JSON requested and validated with Pydantic, malformed output repaired, partial results salvaged, bad results never cached, vision calls retried once. A 40 RPM limit shaped the design: 12 calls instead of 108, calls serialised with a semaphore, results streamed to the UI over server-sent events and cached by input hash. GitHub: https://github.com/Arishtanemi4/wingman.git
- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I designed the RAG-based AI assistant and worked on the scoring algorithm and data preprocessing for a tool, built as FastAPI microservices with a React and TypeScript frontend, that ranks cancer cell lines by target inclusion and exclusion genes and shows the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project; multi-phase study comparing CNN, ResNet and ViT encoders with LSTM and attention decoders for generating natural-language image descriptions on synthetic datasets, benchmarked against pre-trained vision-language models.) GitHub: https://github.com/Arishtanemi4/ata-team30.git

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker), Azure OpenAI, LlamaIndex, Docker, Jenkins, GitLab, Git, Nginx, Linux, Power BI, VS Code, Claude Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic), SQL, R (dplyr, tidyverse, ggplot2), TypeScript (React frontend, CellLineSelector), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite